DUCKIE BIN

OUR PITCH

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PROBLEMS

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1

Overflowing trash

Cleaners waste time emptying half-empty bins while missing the full ones

2

Inefficient Sorting

Lack of data and user error cause massive recycling contamination

3

Unhygenic

Touching bin lids spreads germs in sensitive areas like hospitals and kitchens

SOLUTIONS

Smart Fill Detection

Alerts staff only when the bin actually reaches 90% capacity

Material Detection

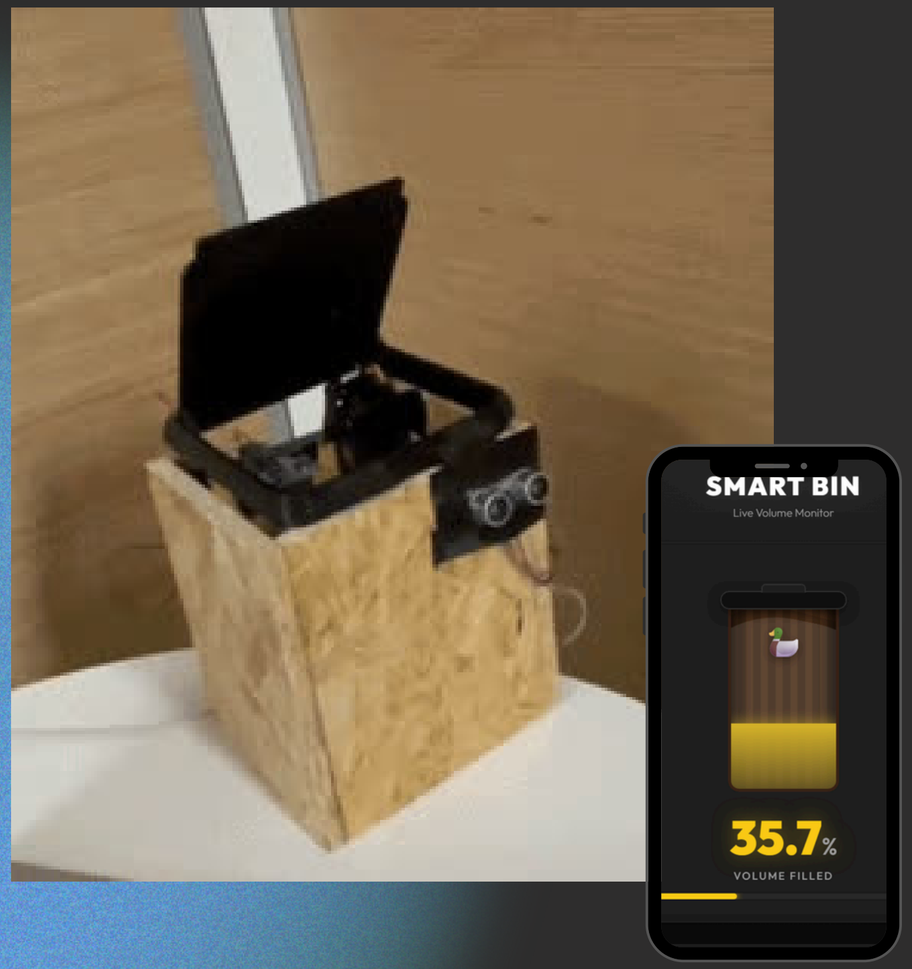
Instantly recognizes materials to guide users and track waste data

Touchless Automation

Motion-sensor lids remove physical contact and improve accessibility

SOLUTIONS

PRODUCT NEW PRODUCT



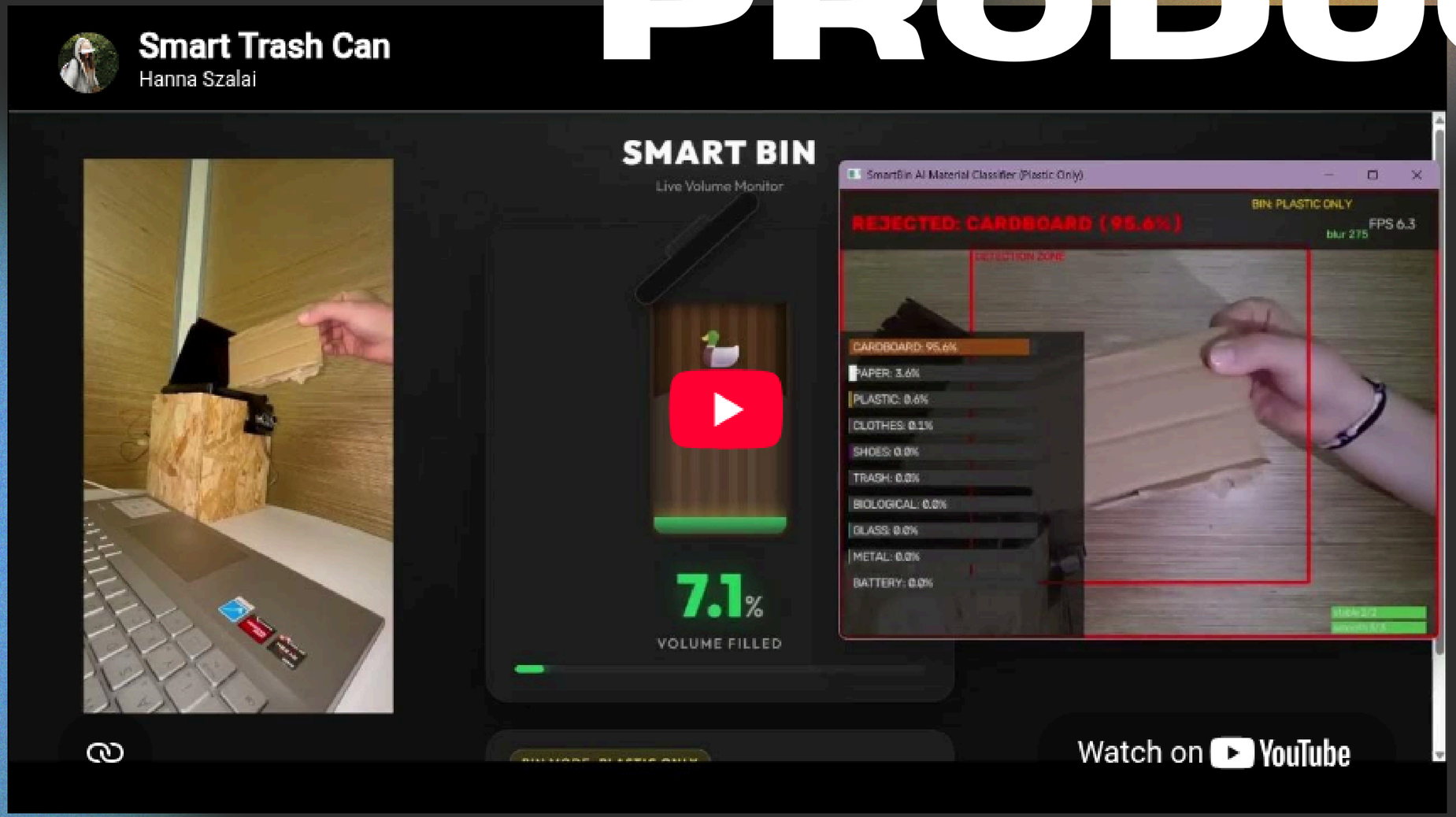
Smart Trash Bin : DuckieBin

Built from standard, replaceable parts for cheap and easy maintenance

Cameras point down into the bin and process images locally without storing them

Shifts waste management from blind fixed schedules to live data

PRODUCT NEW PRODUCT



MARKET

Student Housing

Targeting LOAS & HOAS (~3,000 apartments) where high international student rotation causes chronic sorting errors

Hospitals

Scaling to hospitals, universities, and corporate offices facing strict mandates

Airports

Smart routing cuts waste collection mileage by 20% and CO2 emissions by 18%

TARGET MARKET

Source - IoT-Based Smart Waste Management System: A Solution for Urban Sustainability (Alsayaydeh et al., 2025)

ADVANTAGES

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Labor Savings

Right-time emptying means fewer wasted collection trips across the facility

Less Contamination

Immediate visual feedback flags wrong items before they ruin a recycling batch

Easy Maintenance

Self-checking diagnostics alert staff if a sensor is blocked or a battery is low

FUTURE ROADMAP

NOW

SMART LID
FULLNESS SENSOR
AI DETECTION

FUTURE

SOLAR POWER
CLOUD DASHBOARD
WEIGHT SENSOR
MULTI-COMPARTMENT

THANKS

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